Comparitive Analysis of Time-Series Models for Hourly Electricity Consumption

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Abstract—The rapid rate of development of energy forecasting techniques has increased the necessity of accurate and effective power management. The power demand pattern varies a lot during the day as a result of the patterns of usage seen and the environmental conditions. The paper will compare the analysis of a number of time-series forecasting models that will be utilized to predict hourly energy consumption in electrical grids. The models tested are Long Short-Term Memory (LSTM), Gradient Boosting Machine (GBM), Gated Recurrent Unit (GRU), Hybrid LSTM-GBM ensemble, and Random Forest (RF). New York Independent System Operator is seasonally accessible and publicly accessible hourly load information are experimented with standard measures of performance in a model, Mean Absolute Error (MAE), Root Mean Square error (RMSE) and Mean Absolute Percentage error (MAPE). The findings indicate that deep learning and hybrid models show better functioning than the classical statistical models to find out the temporal relationships within the energy consumption data. The study provides the hands-on data to adopt smart energy forecasting in the electrical grids management systems.

Keywords—power consumption; time series forecasting; LSTM; GRU; GBM; RF; hybrid model; energy management

I. Introduction

The increase in electricity demand in contemporary power grids has altered the way people are considering the way power is to be managed. The amount and complexity of power consumption data has grown a lot because of the rise of connected infrastructure and distributed energy systems. Industry experts say that by 2025, the global energy management market will be worth more than $170 billion [1]. There are opportunities and problems that come with this rapid growth. As an example, grid operators and energy providers should have improved tools to precisely estimate demand and manage loads [2]. The consumption of electricity in the power grids varies continuously and is time-organized. Demand varies during the day, week and year, and varies depending on the way connected systems and end-users utilize it [3]. Such powerful time-related dependencies allow one to perform time series analysis with hourly power consumption and machine learning to perform predictions with ease [4]. Precise hourly energy forecasting can be of great assistance to grid operators and energy systems. First, it allows utility companies to control demand on the customer end which allows them to maximize load allocation at periods of low demand [2]. Second, it can be used to control peak loads by anticipating the time when demand may increase, thus maintaining stability in the grid. Third, with grids powered by a renewable source such as solar and wind, precise forecasting can ensure that supply and demand are more closely aligned, reducing curtailment [5]. Fourth, predictive energy data is utilized in intelligent grid automation and energy dispatch systems to automatically balance the supply and demand.

Over the past ten years, time series forecasting of energy systems has changed a lot. The former procedures were based on classical statistical algorithms like Auto regressive integrated moving annually (ARIMA) and Seasonal autoregressive integrated moving averages (SARIMA). These algorithms are effective in modeling linear time but they are unable to capture the non-linearity in the actual grid load data [6][7]. These statistical methods have been found to have shortcomings, leading to the adoption of machine learning methods, such as Support Vector Machines (SVM), Random Forest (RF), and Gradient Boosting Machines (GBM). These methods of ensemble were shown to be better at representing non-linear and complicated relationships in energy data, but are not necessarily sequential time dependencies [8]. Deep learning has transformed the paradigm of time series forecasting. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are two of the most advanced types of Recurrent Neural Networks (RNN). They are made to find long-term temporal dependencies in sequential data [9][10]. LSTM networks get around the problem of vanishing gradient that makes regular RNNs less effective. They can also learn patterns in energy usage sequences that last for hours or even days. GRU networks are a good choice because they are fast and have similar predictive power [11]. Later hybrid architecture In more recent work, hybrid architectures, or those integrating deep learning models and ensemble techniques, have demonstrated promise concerning state-of-the-art accuracy and robustness [12][13]. Although there has been a lot of research on load forecasting, there are still a number of gaps in the literature. Most current literature concentrates on grid-scale or business building energy prediction, and relatively less is done to concentrate on the finer details of hourly granular hourly grid-level consumption [14]. Also, extensive comparative analyses of the heterogeneous model families: classical machine learning, deep learning, and hybrid models on the same electricity load dataset is limited [15]. It is because of this that practitioners will have a hard time making informed model selection decisions of the electricity load forecasting applications. The paper evaluates these gaps by performing a systematic comparison of five forecasting models, which include LSTM, GBM, GRU, a Hybrid LSTM-GBM model, and the Random Forest on hourly electricity consumption data, obtained through New York Independent System Operator (NYISO) public dataset [16]. The NYISO data is data on actual high-resolution power demand, which is real in both real-world electricity consumption patterns across multiple grid zones. In this paper, the most appropriate model to use is identified and can be applied to make hourly load predictions by performing intensive preprocessing, feature engineering, and model assessment metrics using MAE, RMSE, and MAPE.

The rest of the paper will be structured in the following. Section II introduces a literature review of the related work on the topic of energy consumption forecasting. Section III describes the data set and methodology of preprocessing data. The results and discussion are provided in section IV. Section V provides the future of this research and subsequently it contains references.

II. Literature Survey

Much of the studies in this area have been on predicting energy use at various levels and using various types of models. This section discusses valuable input that will be used in the design of the current study.

A. *Statistical Models*

Initially, early forecasting of the time series relied heavily on statistical time series models. The ARIMA model was applied to predict power consumption by Gopikrishna et al. [6], thus establishing a precedent of the statistical forecasting techniques. This was furthered by Caw-it and Talirongan [17] who investigated the prediction of household electricity by ARIMA with seasonal decomposition and found that it was sufficiently predictive at the weekly level, although less so at the hourly level. Jain et al. [7] performed a general literature review regarding time series analysis methods of electricity consumption prediction, highlighting the value of ARIMA models in predicting stable, linear demand trends but stating that they are incapable of modeling nonlinear dynamics of real-world energy consumption data.

B. *Machine Learning Models*

The shortcomings of purely statistical models led to a consideration of machine learning techniques. Banga et al. [18] have suggested stacked ensemble machine learning models to predict electricity by demonstrating that a combination of the predictions of several learners is better than a single model. Alsalem [19] presented a hybrid time series forecasting system that incorporates fuzzy clustering, with machine learning, with a much higher prediction accuracy through consumption profile segmentation followed by the application of model specific learners. A comparative study of several forecasting models such as SVM, Random Forest, and Gradient Boosting was done by Bilal et al, [8], which found that the tree-based ensemble forecasting methods provide a good balance between prediction accuracy and computing cost on real-world energy data.

C. *Deep Learning Models*

Deep learning models show greater capabilities in the process of capturing temporal dependencies in energy data. Qureshi et al. [9] used LSTM networks to predict electricity demand and showed it to be highly effective compared to ARIMA baselines (based on RMSE). Ullah et al. [13] showed a CNN-LSTM hybrid framework of load forecasting in the short term by extracting spatial features using a convolutional layer and then sent them through an LSTM to model time. Wang et al. [20] explored how the characteristics of temperature affect LSTM-based electricity forecasting, indicating that exogenous variables can effectively improve the model. Ucar and Kaygusuz [11] compared various deep learning models with their various optimization strategies and discovered thatGRU networks with an adaptive learning rate calendars exhibit competitive outcomes with fewer parameters than LSTM.

D. *Hybrid Models*

There has been a tremendous interest in hybrid models that integrate several modeling paradigms. The hybrid energy prediction deep learning architecture suggested by Hammou Ou Ali et al. [12] was used to predict the energy consumption by using convolutional features and recurent time models, and it outperforms the current state-of-the-art on a benchmark. In Allehyani [21], a hybrid CNN-LSTM model with TimeGAN-based data augmentation was established, which exhibits better generalization in the conditions of data lack. In their proposed building energy prediction, Ngo et al. [22] used a SARIMA-SVR hybrid, which demonstrated that a statistical seasonal decomposition approach used along with a kernel-based regressor provided robust predictions. The proposed hybrid machine learning model suggested by Parizad et al. [23] to jointly forecast energy demand and electricity price at the same time has an advantage in the multi-task learning used in smart energy usage. Data privacy concerns in energy forecasting have been an incentive to federated learning. A federated transfer learning model of smart household energy prediction proposed by Rizwan et al. [24] maintains the privacy of the user and delivers accuracy at the level of the centralized one. Such direction is a new frontier but not a part of the current study which narrows down to centralized model comparison on publicly available dataset.

Regardless of the scope of literature available, there are still some significant deficiencies. Most research is dedicated to power or grid-scale data, and less research is done to consider hourly grid-level consumption forecasting at a granular scale. There exists little literature on comparative studies comparing heterogeneous model families, i.e. both ensembles of machine learning and deep learning architectures, on the same electricity load dataset based on consistent evaluation protocols. Additionally, hybrid frameworks that involve GBM and LSTM specifically to do hourly electricity load forecasting have not been fully ventured. The paper fills these gaps and analytically compares LSTM, GBM, GRU, a Hybrid LSTM-GBM model, and the Random Forest model on the NYISO hourly electricity data under the same experimental settings.

III. Data Description

A. Dataset Source

The data employed in the current research is based on the New York Independent System Operator (NYISO), a publicly available database of real-time and historical data of state New York electricity demand [16]. NYISO controls the electric grid and wholesale electricity markets of New York State and releases hourly data on electricity loads aggregated across several zones such as the New York city, Long Island, and upstate among others. The NYISO dataset is accessible over the NYISO site and is extensively utilized in energy prediction studies thanks to its dependability and timeliness and granularity. On the basis of this research, hourly integrated actual load real time data in NYISO over the course of several years was downloaded in order to have enough volume of data to train, validate, and test the time series models. The data is aggregated real-time electricity demand (in megawatt-hours (MWh)) at hourly intervals. The NYISO data is represented by the regional grid-level demand aggregated across multiple zones. The time structure is on an hourly basis, seasonally stable, and exhibits diurnal and seasonal patterns representative of real-world electricity consumption [25] and, thus, is an excellent proxy to test time series forecasting models in that respect.

B. Dataset Characteristics

The raw data is a collection of time-stamped data based on the hourly load value of electricity in selected load zones by NYISO. The main structural characteristics of the dataset are the following. In each record, there is a timestamp field that indicates the date and hour in which it was recorded, a zone with a number to indicate geographical area, and a load value in MWh. The data set has high temporal dynamics on a variety of scales, a diurnal cycle due to human activity, a weekly cycle due to the difference between weekdays and weekends in consumption activity and a seasonal cycle which is due to heating and cooling demand which vary with temperature. These multi-scale periodic structures permit the dataset to be highly applicable in assessing the functionality of various models of time series to model time dependencies at different granularities. The data set covers a number of years of hourly observations, which makes tens of thousands of observations. The present volume can be used to train deep learning models as LSTM or GRU, which often require large sequence datasets in order to learn substantial temporal representations. No direct sub-metering data is present in the dataset, but the aggregate hourly load signal does provide adequate data to formulate overall consumption patterns to conduct the forecasting exercise in the current study.

C. Data Preprocessing

To guarantee the quality and appropriateness of the data in time series modelling, data preprocessing was carried out in a few consecutive phases. The raw CSV files that had been downloaded on the NYISO portal were initially thrashed and condensed into a single unified dataframe. The pandas library has been used to convert timestamp columns in string format to Python datetime objects, as that way data can be properly indexed by time and resampled using time-related operations. The plain issue of missing values was handled by a mixture of short interval (gap) linear interpolation and a one-off situation (a single record) fill forward imputation. The duplicate timestamps, which sometimes occur when changing the daylight saving times, were solved by averaging the duplicate records. An outlier procedure involved the downloading which centered a 168 hour (a one-week) rolling z-score to identify load values outside of three standard deviations of the means and was applied to the hourly load data: outliers were labeled and substituted with linearly interpolated values based on the adjacent measurements.

The feature engineering was done in order to deepen the input feature space on top of the original load values. The temporal features of the timestamp index are the hour of the day (0-23), day of the week (0-6), day of the month (1-31), month of the year (1-12) and a weekend (0-1) discerner. Lags with values of previous consumption at lag 1, 2, 3, 6, 12, 24, and 168 hours were calculated in order to give the models clear historical context. Rolling statistics such as 24-hour rolling mean and 24-hour rolling standard deviation had been calculated and fitted as features. In the case of deep learning models (LSTM and GRU), the input data was organized as three-dimensional tensors with a shape of (samples, time steps, features), where a sliding window mode was used with a lookback window of 24 hours to estimate the consumption value in the following hour. In case of machine learning models (GBM, Random Forest), the lagged features and rolling statistics were put in a flat two-dimensional feature matrix. The min-max scaling was done to normalize all feature values to the range [0, 1] to allow easy training of gradient-based deep learning networks.

D. Dataset Split

A chronological split was used to split the dataset into training, validation and test subsets to ensure that time sequence of data is considered and avoids data leakage. Training, 15% to training, and validation (hyperparameter tuning and early stopping) took about 70 percent of data, and testing the remaining 15 percent. The test set will be made up of the latest section of data, and this allows an assessment of model-performance on truly unseen future data. The reason behind this bifurcation approach is conventional practice in the evaluation of time series forecasting and provides a reasonably fair comparison of all five of these models.

E. Exploratory Data Analysis

Exploratory data analysis (EDA) was performed to describe the statistical characteristics and the time distribution of the analyzed data. Hourly plots of load data indicate a definite diurnal trend with a peak load occurring in the late afternoon and early evening hours (around 17:00-20:00) and minimum occurring at the early time of the day (around 03:00-05:00). Patterns indicate that the consumption is usually low on weekends as compared to weekdays. The seasonal analysis shows that demand reaches peak in the summer (because of the air conditioning load) and the second peak is expected in the winter (because of the electric heating).

The presence of the significant autocorrelation at the lags of 1, 24, and 168 hours is confirmed by the use of the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots, therefore, the presence of a lag at 1, 24 and 168 hours is justified. Distribution analysis reveals that the values of hourly loads are normally distributed with minimal skewness towards the right indicating the odd spike in demand during extreme weather conditions. The results of these EDA were used to make decisions on feature engineering, as well as the choice of a suitable model architecture to be used in the forecasting experiments that will be presented below.

IV. Result and Discussion

This section shows the experimental findings through the training and testing of five forecasting models, namely, GRU, LSTM, Random Forest, GBM, as well as the proposed model of Hybrid LSTM+GBM model when applied on NYISO hourly the electricity load data set. Four standardized measures were used to evaluate each model, with roots mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination, (R^2). A smaller RMSE, MAE and MAPE means that forecasts are more accurate and the forecasting power based on a higher R^2 is achieved. Fig. 1-5 demonstrate comparisons between the predicted load curves and the actual load curves of each model within a 500 hrs test window. The models are listed from weakest to strongest, so you can see the differences before we talk about the numbers.

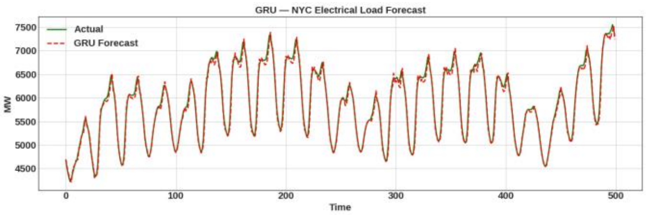

Fig. 1. Forecast vs. Actual load using GRU

The GRU captures the overall daily trend, but it doesn't always follow it exactly during peak demand hours. This makes it the model with the highest RMSE among all the ones we looked at.

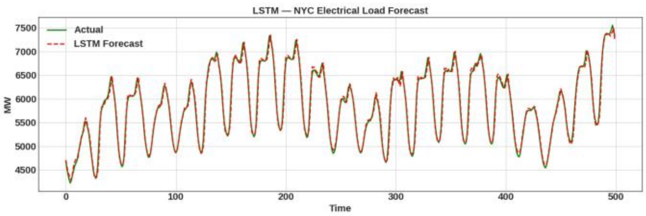

Fig. 2. Forecast vs. Actual load using LSTM

The LSTM does a better job of keeping track of load peaks and troughs than the GRU. This is because its long-term memory cells are better at modeling patterns of energy use over time.

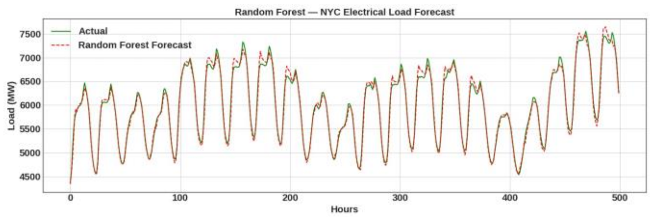

Fig. 3. Forecast vs. Actual load using RF

The Random Forest does a good job of following the general trend, but it clearly under-predicts at a few peaks, which is in line with its high MAPE of 8.87%.

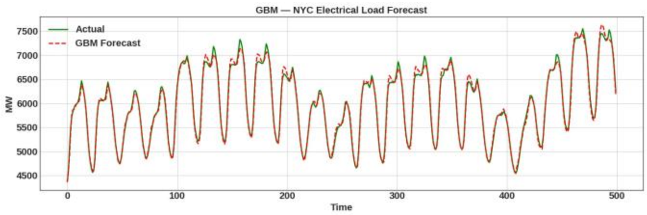

Fig. 4. Forecast vs. Actual load using GBM

The GBM forecast is very close to the actual load signal, with peaks that are more closely aligned than those of Random Forest. It has a MAPE of 0.75% and an RMSE of 62.70 MW.

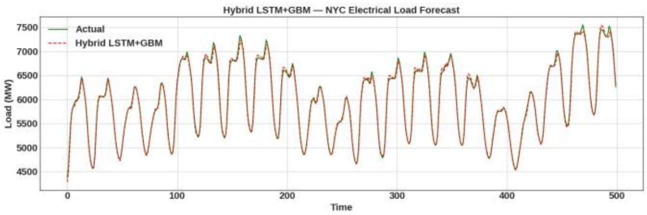

Fig. 5. Forecast vs. Actual load using Hybrid LSTM + GBM

The Hybrid model achieves the best alignment with the actual load curve across the full test window, yielding the best overall performance with an RMSE, 52.50 MW and R^2, 0.9979.

TABLE I. Performance Comparison of Forecasting Models

Model	RMSE	MAE	MAPE (%)	R^2
GRU	87.98	63.85	1.1024	0.9941
LSTM	78.34	58.22	1.0264	0.9954
Random Forest	72.94	51.04	8.8658	0.9960
GBM	62.70	43.97	0.7506	0.9954
Hybrid LSTM + GBM	**52.50**	**35.43**	**0.6043**	**0.9979**

The performance of each of the five models is outlined in Table I. According to its results, it is evident that the accuracy has improved gradually in the standalone GRU model up to the Hybrid LSTM+GBM model. GRU made an RMSE of 87.98 MW and a MAPE of 1.10 that is the baseline of deep learning performance. The LSTM showed a margin of this by 78.34 MW RMSE and 1.03 MW MAPE on sequential load data testing, indicating that gated memory architectures do outperform. The RMSE and R^2 of Random Forest were 72.94 MW and 0.9960 respectively with a very high MAPE of 8.87% implying sensitivity to relative errors during low load conditions.

GBM model provided a better result of 62.70 MW and a MAPE of 0.75%. It indicates that gradient-boosted trees are useful in identifying the non-linear relationships in real-world electricity load patterns using lag features as inputs. Hybrid LSTM+GBM model obtained the highest performance in all the measures with an RMSE of 52.50 MW, MAE of 35.43 MW, MAPE of 0.60, and an R^2 of 0.9979. This is a 40.3 percent lower RMSE than the baseline GRU and indicates how well LSTM and GBM are sequentially pattern and residual error correcting, respectively.

This superior result of the Hybrid LSTM+GBM model could have been explained by the fact that the structure of the Hybrid comprises two phases: LSTM component learns the time structure and superior periodic patterns of the load sequence, and the GBM component could address the residual error made by the LSTM, using engineered lag features and calendar covariates. High R^2 of 0.9979 attests to the fact that hybrid model captures more than 99.7% of variance in the unknown hourly load data, thus it is very appropriate to be applied in intelligent energy management and grid forecasting systems.

V. Conclusion

Finally, the experimental findings indicate that the Hybrid LSTM+GBM model (RMSE 52.50 MW, MAE 35.43 MW, MAPE 0.60, R^2 0.9979) shows the highest level of performances compared to all five models that were discussed, and this is a reduction in RMSE by 40.3 percent experimentally. Two-stage architecture, in which LSTM is used to collect load patterns over time and GBM is used to fix possible residual errors, is extremely useful in hourly power consumption prediction and is the most appropriate candidate to be deployed in an intelligent energy management system.

The direction of the future work will comply with the concept of making the greatest gains in improving the model generalizability through incorporating multi-region datasets and addressing transfer learning in data scarcity conditions. Exogenous weather variables and sub-metering data integration are also planned to enhance the granularity of forecasting to be more precise and manage the demand side. Lastly, studies will look into the creation of real-time, edge-deployable learning models and the combination of load forecasting and renewable energy optimization in support of holistic and action reacting energy management.

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